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3 February 1981
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3 February 1976

MILITARY SPECIFICATION

INDICATOR, PRESSURE

This specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the design, manufacture, inspection and testing of pressure indicators.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-N-411 Nitrogen, Technical.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, N. J. 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

DOD-D-1000	Drawing, Engineering and Associated Lists.
MIL-C-5501	Cap and Plug, Protective, Dust and Moisture Seal.
MIL-B-22020	Bag, Transparent, Flexible, Sealable, Volatile Corrosion Inhibitor, Treated.

STANDARDS

MILITARY

MIL-STD-109	Quality Assurance Terms and Definitions.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of.
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of.
MIL-STD-810	Environmental Test Methods.
MIL-STD-831	Test Reports, Preparation of.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

DL 58A164D556	Receiver Assembly, Nitrogen.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4 and 6.3).

3.1.1 First article sample. Unless otherwise specified in the contract or purchase order (see 6.2.1), a first article sample of four pressure indicators shall be delivered to the testing activity designated in the contract or purchase order (see 6.2.1). The first article sample shall be manufactured using the same methods, materials, processes and procedures proposed for production. Any production prior to acceptance of the first article sample is at the contractor's risk.

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3.2 Selection and quality verification of parts and materials. Quality verified parts (see 3.2.1) and materials from approved sources (see 3.2.2) shall be used in the fabrication of the end item. The contractor shall prepare test reports of conformance to applicable requirements.

3.2.1 Approval of parts and materials. Parts and materials proposed for use in the fabrication of the end item that are not specified in detail herein or in applicable referenced documents shall be selected from other documents in the order of precedence specified in MIL-STD-143 and shall be approved by the procuring activity prior to use. Approval shall be predicated upon source (vendor) approval and verification of part quality by test.

3.2.2 Source approval. The contractor shall be responsible for ascertaining that each proposed source for parts and materials maintains a quality control program to ensure consistent delivery of products conforming to applicable requirements. When parts or materials are fabricated by the contractor, the area used for that purpose shall be considered the same as an outside vendor insofar as source approval is concerned. Source approvals by the contractor shall be subject to concurrence by the procuring activity. Source approval will be granted by the procuring activity upon preparation of acceptable evidence of any of the following:

a. Formal on-site survey of the vendor's manufacturing facilities.

b. The source has been previously approved, and evidence is available to confirm that acceptable product quality has been maintained.

c. Other evidence of acceptable product quality is available.

3.2.3 Verification of part quality. Quality verification shall be demonstrated by successful testing of the part to levels of environmental, electrical, and physical stresses equal to or greater than those which will be encountered in the system and as required by the part specification. Part quality may be demonstrated by any of the following:

a. When the part is manufactured and qualified to a military specification which includes requirements equal to or more severe than those of the end item application.

b. When the part is manufactured and qualified to a military specification not including all requirements of the application, and additional tests for such requirements have been performed.

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c. When the part is not qualified to a military specification, but all tests appropriate to the application have been performed.

3.2.4 Approved source/quality verified parts list (AS/QVPL). An AS/QVPL shall be prepared by the contractor for all subassemblies and parts for approval by the procuring activity. All entries on the AS/QVPL shall be fully substantiated by the preparation of data to satisfy the requirements of source approval and part test approval as specified herein. Reports of all quality verification testing shall conform to the requirements of MIL-STD-831. The contractor may solicit from the procuring activity information as to the status of quality verified parts and approved sources which may have been previously developed. Neither part environmental tests nor part environmental test reports are required in those instances where vendors are selected from current AS/QVPLs of other programs. Entries on the AS/QVPL shall identify the vendor and the product which is proposed for use in the pressure indicator by the contractor. In the event several sources have previously been approved for the application, the contractor shall specify the source of the parts to be used. Each item listed shall be supported in accordance with the provisions of this section. This listing does not supplant or modify the applicable documentation or other Government Qualified Parts Listing.

3.2.4.1 Conditions governing the quality verification of parts. When the contractor proposes to demonstrate the quality verification of parts, as provided herein, the following provisions shall apply:

a. Test may be performed by the contractor, utilizing his own facilities, the facility of the manufacturer of the part or by any other suitable facility selected by the contractor. In any event, the facilities shall be approved by the contractor prior to the performance of any test. This information shall be prepared for approval by the procuring activity.

b. Prior to the performance of any test, a detailed test procedure shall be prepared by the contractor and approved by the procuring activity. The procedure shall identify, as a minimum, the step by step functions to be performed and the test equipment to be used, and shall provide for full conformance with the provisions of the applicable documentation. The Government reserves the right to review any of the individual test procedures at its discretion.

c. Quality verification testing may be monitored by a representative of the procuring activity.

3.2.4.2 Changes to the AS/QVPL. The contractor may develop alternative sources for any part, subject to the provisions herein. Proposals by the contractor for part substitution or changes to the AS/QVPL shall include proof

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that the proposed substitute part will meet the requirements of 3.2.2 and 3.2.3. The AS/QVPL shall be revised and updated as applicable to include approved changes.

3.2.4.3 Removal from the AS/QVPL. Evidence of poor quality may be cause for suspension or removal of a source from the AS/QVPL at the option of the contractor or the procuring activity. The contractor shall develop quality verified sources for those parts for which the approved source has been removed. Quality reverification shall be required under any of the following conditions unless otherwise directed by the procuring activity:

a. When the part has been modified by a change in design, materials, manufacturing process, manufacturing inspection procedures, or manufacturing facilities. The procuring activity shall be notified prior to delivery of any parts incorporating any such changes and be provided quantitative evidence of the effect of such changes on the performance or other characteristics of the pressure indicator.

b. When there is evidence that the quality of the part has not been maintained. This evidence may be in the form of accumulated failure reports which indicated that the quality of the part has deteriorated, or there is a failure trend.

c. When the manufacturer has not produced the item for 6 or more months.

3.2.5 Part conformance provisions. When a drawing or specification for a part invokes by reference the provisions of a military specification, total conformance to the provisions of the referenced specification shall be required. Total conformance is defined as conformance to the stated performance and product characteristics, and compliance with all quality assurance provisions, including qualification, quality verification, and quality conformance testing.

3.3 Design and construction. The design and construction of the pressure indicators shall be in accordance with the applicable drawings listed on DL 58A164D556 and as specified herein. The contractor shall notify the procuring activity of any changes in design and construction after the charging valve has been accepted by the procuring activity.

3.3.1 Materials. Materials shall be selected on the basis of strength, workability and compatibility with other materials in the item so as to minimize galvanic action, corrosion, etc.

3.3.2 Finishes. Finishes shall be applied to all materials, whenever applicable and shall be for the purpose intended, such as corrosion resistance,

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increased durability, etc. External surfaces shall be in accordance with the applicable drawings listed on DL 58A164D556.

3.3.3 Interchangeability. All pressure indicators having the same part number shall be internally and externally identical. Changes in part numbers shall be governed by drawing number requirements of DOD-D-1000.

3.3.4 Identification of parts. All pressure indicators shall be marked as specified on the applicable drawings listed on DL 58A164D556.

3.3.5 Dimensions. The pressure indicator envelope dimensions shall be in accordance with the applicable drawing for the pressure indicator being procured.

3.4 Performance. The pressure indicator, when constructed in accordance with the applicable drawings listed on DL 58A164D556, shall meet the requirements as specified herein.

3.4.1 Cleanliness. The pressure indicator shall be free of all water, oils, grease and foreign matter exceeding 0.002 inches in size on all surfaces.

3.4.2 Leakage. The pressure indicator shall not leak when subjected to pressures between 200 and 3800 pounds per square inch gage (psig).

3.4.3 Operating pressure. The pressure indicator shall operate satisfactorily after being subjected to 5000 psig.

3.4.4 Environmental. The pressure indicator shall meet the following environmental requirements of MIL-STD-810, unless otherwise specified herein.

3.4.4.1 High temperature. The pressure indicator shall operate satisfactorily after the test of MIL-STD-810, Method 501, Procedure I, except the temperature shall be 121°C (250°F).

3.4.4.2 Low temperature. The pressure indicator shall operate satisfactorily after the test of MIL-STD-810, Method 502, Procedure I for a period of four hours.

3.4.4.3 Temperature shock. The pressure indicator shall operate satisfactorily after the test of MIL-STD-810, Method 503, Procedure I.

3.4.4.4 Dust. The pressure indicator shall operate satisfactorily after the test of MIL-STD-810, Method 510, Procedure I.

3.4.4.5 Salt fog. The pressure indicator shall operate satisfactorily after the test of MIL-STD-810, Method 509, Procedure I, except the duration shall be 50 hours.

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3.4.4.6 Humidity. The pressure indicator shall operate satisfactorily after the test of MIL-STD-810, Method 507, Procedure I, except four 24 hour cycles shall be substituted for the ten 24 hour cycles.

3.4.4.7 Vibration. The pressure indicator shall operate satisfactorily after exposure to the vibration tests specified herein.

3.4.4.8 Shock. The pressure indicator shall operate satisfactorily after the test of MIL-STD-810, Method 516, Figure 516-2, except the acceleration shall be 25 g's and shall reach peak value in not more than 0.600 second and not less than 0.015 second.

3.4.5 Service life. The pressure indicator shall operate satisfactorily and show no damage after begin subjected to cycling tests as specified herein.

3.4.6 Proof pressure. The pressure indicator shall operate satisfactorily and show no indication of failure when pressurized to $5500\ ^{+0}_{-100}$ psig for three minutes.

3.4.7 Burst pressure. The pressure indicator shall operate satisfactorily and show no indication of rupture or internal leakage after being pressurized to 6700 ±100 psig for a minimum duration of three minutes.

3.5 Precedence. The selection of specifications and standards for use in the design and development shall be in accordance with the requirements of MIL-STD-143. When the requirements, this specification, and referenced documents are in conflict, the following precedence shall apply:

a. This specification. This specification shall have precedence over all referenced documents.

b. Referenced documents. Any documents referenced in this specification shall have precedence over all subsidiary documents referenced therein.

3.6 Workmanship. The workmanship displayed in fabrication and assembly of the pressure indicator shall be such as to assure, within design limitations, the ability of the pressure indicator to meet its performance requirements under all conditions specified herein. Particular attention shall be paid to neatness and thoroughness of assembly, marking of parts and assembly, plating, painting, machining, welding, brazing and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in the approved first article sample, subject to any qualification stated in the Government's notice of approval, shall determine the requirements of the remainder of the contract relative to workmanship.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure that the supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection of the pressure indicator shall be classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed under the following conditions and tolerances.

a. Room temperature. Room ambient of 25 ±10°C (77 ±18°F) indicated.

b. Test temperature. All parts of the test item shall be stabilized at the specified temperature ±2.8°C (±5°F) indicated prior to conducting any operational tests. Unless otherwise specified herein, temperature stabilization will have been attained when the indicated temperature of the surface of the largest mass of the test item does not change more than ±2.8°C (±5°F) in a period of one hour.

c. Humidity. Room ambient to 95 percent relative humidity.

d. Test gas. The test gas shall be nitrogen in accordance with BB-N-411, Type I, Class I, Grade A.

e. Test fixture. A high-strength, corrosion-resistant fixture shall be utilized for mating with the pressure indicator port. The fixture shall have a pressure indicator port in accordance with the applicable drawing listed on DL 58A164D556 and an open port for gas transfer. The pressure indicator shall be torqued to the fixture. The test fixture shall be used for all tests of 4.6.3, 4.6.4, 4.6.5 and 4.6.6.

f. Torque requirements. The installation torque for the pressure indicator to the receiver assembly or test fixture shall be 100 ±25 inch pounds.

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4.4 First article inspection. First article inspection shall consist of the tests outlined in Table I (see 6.2.1). The procuring activity reserves the right to delete tests and alter the test sequence of Table I. First article approval is only valid on the contract or purchase order under which it is granted, unless extended by the Government to other contracts or purchase orders.

4.4.1 Pressure indicator samples. First article pressure indicator samples shall consist of the first four pressure indicators manufactured on the contract or purchase order. The pressure indicators shall be permanently labeled for ease of identification and shall be produced with equipment and procedures that will be used for the production of all the pressure indicators designated in the contract or purchase order. Prior to submission, the contractor shall inspect the samples to the degree necessary to assure that they conform to the requirements of the contract and prepare a first article test report of this inspection with the samples including statements of findings and certificates of conformance for materials, cleanliness and tests A through E of Table I.

4.5 Quality conformance inspections. Acceptance of pressure indicators shall be on an inspection lot basis. Quality conformance inspections shall consist of the following tests:

a. Individual tests (see Table II, Group 1).

b. Sampling tests (see Table II, Groups 1 and 2).

4.5.1 Individual tests. Individual tests are those tests conducted by the contractor on each pressure indicator submitted for acceptance under the contract or purchase order. Individual tests for pressure indicators are shown in Table II, Group 1. Failure to pass these tests shall be cause for rejection.

4.5.2 Sampling tests. A random sampling of pressure indicators shall be selected by the procuring activity from each inspection lot in accordance with Table III and shall be subjected to the tests listed in Table II, Groups 1 and 2. Group 1 tests shall be conducted prior to Group 2 tests. Failure to pass these tests shall be cause for rejection of the lot. The procuring activity reserves the right to delete tests and alter the test sequence of Table II (see 6.2.1). In the event any sample representing an inspection lot fails to meet the inspection and test requirements outlined herein, the contractor, at the discretion of the procuring activity, may have the opportunity of reworking the entire affected lot. The procuring activity shall determine which inspections and tests of this specification are to be performed on the reworked inspection lot samples. The number of test samples shall be at the discretion of the procuring activity (see 6.2.1).

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TABLE I. First article inspections - schedule and sequence[1/].

Test	Test title	Test requirement paragraph	Test method paragraph	Sample			
				A	B	C	D
A	Cleanliness	3.4.1	4.6.1	1	1	1	1
B	Inspection	3.1.2	4.6.2	2	2	2	2
C	Proof pressure	3.4.5	4.6.3.1	3	3	3	3
D	Leakage	3.4.2	4.6.3.2	4	4	4	4
E	Operating pressure	3.4.3	4.6.3.3	5	5	5	5
F	High temperature	3.4.4.1	4.6.4.1	6	X	6	X
G	Low temperature	3.4.4.2	4.6.4.2	7	X	7	X
H	Temperature shock	3.4.4.3	4.6.4.3	X	6	X	6
I	Dust	3.4.4.4	4.6.4.4	X	7	X	7
J	Salt fog	3.4.4.5	4.6.4.5	X	8	X	8
K	Humidity	3.4.4.6	4.6.4.6	X	9	X	9
L	Vibration	3.4.4.7	4.6.4.7	8	X	8	X
M	Shock	3.4.4.8	4.6.4.8	9	X	9	X
N	Service life[2/]	3.4.5	4.6.5	X	10	X	10
O	Burst pressure	3.4.7	4.6.6	10	X	10	X

[1/] Test sequence is denoted by the numbers in the sample columns. An X denotes test not required.

[2/] Pressure indicators subjected to life tests shall not be subjected to the burst pressure test.

4.6 Inspection methods.

4.6.1 Cleanliness. The pressure indicator shall be subjected to the cleanliness test detailed herein. The test shall be conducted in a clean, dust-free atmosphere and all equipment shall be thoroughly cleaned. The pressure indicator shall be visually examined for cleanliness and shall be free of all water, oils, grease and foreign matter exceeding 0.002 inches in size on all surfaces internally interfaced with the nitrogen receiver.

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TABLE II. Quality conformance inspections - schedule and sequence[1/].

Tests	Test requirement paragraph	Test method paragraph	Sample			
GROUP 1			ALL SAMPLES			
Cleanliness	3.4.1	4.6.1	1			
Inspection	3.1.2	4.6.2	2			
Proof pressure	3.4.5	4.6.3.1	3			
Leakage	3.4.2	4.6.3.2	4			
Operating pressure	3.4.3	4.6.3.3	5			
GROUP 2[2/]			A	B	C	D
High temperature	3.4.4.1	4.6.4.1	1	X	1	X
Low temperature	3.4.4.2	4.6.4.2	2	X	2	X
Temperature shock	3.4.4.3	4.6.4.3	X	1	X	1
Dust	3.4.4.4	4.6.4.4	X	2	X	2
Salt fog	3.4.4.5	4.6.4.5	X	3	X	3
Humidity	3.4.4.6	4.6.4.6	X	4	X	4
Vibration	3.4.4.7	4.6.4.7	3	X	3	X
Shock	3.4.4.8	4.6.4.8	4	X	4	X
Service life[3/]	3.4.5	4.6.5	X	5	X	5
Burst pressure	3.4.7	4.6.6	5	X	5	X

[1/] Test sequence is denoted by the numbers in the sample columns. An X denotes test not required.
[2/] The procuring activity shall decide which tests of Table II, Group 2, each additional pressure indicator in excess of four shall receive (see 6.2.1).
[3/] Pressure indicators subjected to life tests shall not be subjected to the burst pressure test.

4.6.2 Inspection. The pressure indicator shall be inspected as follows:

4.6.2.1 Individual tests-visual. The pressure indicator shall be inspected to ensure that it has been properly assembled, and that the workmanship is as specified in 3.6.

4.6.2.2 First article and quality conformance tests. The pressure indicator shall be subjected to the examinations and tests of Tables I and II.

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binding or excessive lost motion. The controls shall be designed

TABLE III. Sampling test items.

Lot size	Number of samples to be tested
100 or less	2
101 to 300	3
301 to 500	4
501 or more	4 plus 1 for each additional 500 or fraction thereof.

Failure of the item to pass examinations and tests within the criteria of the sampling plan specified in 4.5.2 shall cause rejection of the lot. The pressure indicator shall be inspected as follows:

a. The pressure indicator shall be thoroughly inspected to determine conformance to all the requirements of the applicable drawings listed on DL 58A164D556 and this specification.

b. The pressure indicator shall be examined to ascertain that the packaging, packing and marking is in accordance with Section 5.

4.6.3 Functional tests. The pressure indicator shall be subjected to the following tests.

4.6.3.1 Proof pressure. The pressure indicator shall be subjected to a proof pressure test. The pressure indicator shall be pressurized to 5500 $^{+0}_{-100}$ psig for a maximum of three minutes. There shall be no indication of failure.

4.6.3.2 Leakage. The pressure indicator shall be checked for leakage with its inlet port pressurized to each of the following pressures for three minutes:

a. 200 psig

b. 1000 psig

c. 2200 psig

d. 3000 psig

e. 3800 psig

There shall be no indication of leakage at any of the pressures between 200 and 3800 psig.

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4.6.3.3 Operating pressure. The pressure indicator shall be subjected to an operating pressure test. The pressure indicator shall be pressurized to the following pressures:

a. 0 psig

b. 225 psig

c. 2200 psig

d. 2800 psig

e. 3500 psig

f. 5000 psig

The pressure indicator dial reading for the 225 psig level shall be approximately 10% of the red scale. The remaining dial readings at each level shall be within ±150 psig of the applied pressure. The 5000 psig pressure level shall be tested only once during the initial operating pressure.

4.6.4 Environmental tests. The pressure indicator shall be subjected to the following environmental tests.

4.6.4.1 High temperature. The pressure indicator shall be subjected to the high temperature test in accordance with MIL-STD-810, Method 501, Procedure I, except the temperature shall be 121°C (250°F). While at temperature conditions, the pressure indicator inlet port shall be pressurized and maintained at 3800 $^{+0}_{-200}$ psig. At the completion of the test, the pressure indicator shall be visually inspected for damage and tested in accordance with 4.6.3.2 and 4.6.3.3.

4.6.4.2 Low temperature. The pressure indicator shall be subjected to the low temperature test in accordance with MIL-STD-810, Method 502, Procedure I for a period of four hours. The storage temperature shall be -54°C (-65°F). While at temperature conditions, the pressure indicator inlet port shall be pressurized and maintained at 3800 ±100 psig. At the completion of the test, the relief valve shall be visually inspected for damage and tested in accordance with 4.6.3.2 and 4.6.3.3.

4.6.4.3 Temperature shock. The pressure indicator shall be subjected to the temperature shock test in accordance with MIL-STD-810, Method 503, Procedure I. Prior to being subjected to temperature conditions, the pressure indicator inlet port shall be pressurized to 3000 ±100 psig. At the completion of the test, the pressure indicator shall be visually inspected for damage and tested in accordance with 4.6.3.2 and 4.6.3.3.

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4.6.4.4 Dust. The pressure indicator shall be subjected to the dust test in accordance with MIL-STD-810, Method 510, Procedure I. While at test conditions, the pressure indicator inlet port shall be pressurized to 225 ±25 psig. At the completion of testing, the pressure indicator shall be visually inspected for damage and tested in accordance with 4.6.3.2 and 4.6.3.3.

4.6.4.5 Salt fog. The pressure indicator shall be subjected to the salt fog test in accordance with MIL-STD-810, Method 509, Procedure I except the duration shall be 50 hours. While at test conditions, the pressure indicator inlet port shall be pressurized to 225 ±25 psig. At the completion of testing, the pressure indicator shall be visually inspected for damage and corrosion, and tested in accordance with 4.6.3.2 and 4.6.3.3.

4.6.4.6 Humidity. The pressure indicator shall be subjected to the humidity test in accordance with MIL-STD-810, Method 507, Procedure I, except four 24-hour cycles shall be substituted for the ten 24-hour cycles. While at test conditions, the pressure indicator inlet port shall be pressurized to 225 ±25 psig. At the completion of testing, the relief valve shall be visually inspected for damage and corrosion, and tested in accordance with 4.6.3.2 and 4.6.3.3.

4.6.4.7 Vibration. The pressure indicator shall be subjected to the following test.

4.6.4.7.1 General. The vibration tests specified herein shall be performed in each of three mutually perpendicular axes in the order shown. All testing shall be performed in one axis prior to testing in the next axis. As indicated herein, the pressure indicator shall be checked for proper operation.

4.6.4.7.2 Pressure indicator operation. While subjected to vibration the pressure indicator inlet port shall be pressurized to 3000 ±100 psig. After each axis of vibration is completed, the pressure indicator shall be visually inspected for damage and tested in accordance with 4.6.3.2 and 4.6.3.3.

4.6.4.7.3 Mounting techniques. The pressure indicator shall be attached to the vibration exciter table by means of the fixture described in 4.3.e. The vibration input shall be measured on the fixture at a point adjacent to the mounting point of the pressure indicator.

4.6.4.7.4 Sinusoidal test level. The pressure indicator shall be subjected to sinusoidal vibration along each of three mutually perpendicular axes. The frequency shall be ±2 percent and the amplitude shall be ±5 percent. The test

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shall consist of sweeping for 60 minutes in each axis from 5 to 2000 Hz at one octave per minute in accordance with Table IV.

TABLE IV. Sinusoidal test level.

Frequency	Displacement or acceleration
5 - 20 Hz	0.100 inch da
20 - 33 Hz	±2.0 g's
33 - 74 Hz	0.036 inch da
74 - 2000 Hz	±10.0 g's

4.6.4.7.5 Random test level. The pressure indicator shall be subjected to a random vibration along each of three mutually perpendicular axes. The instantaneous random vibration acceleration peaks shall be limited to three times the rms acceleration level. The power spectral density of the test control signal shall not deviate from the specified requirements by more than +100 -50 percent (±3 dB) between 20 Hz and 2,000 Hz, except that deviations as large as +300 -75 percent (±6 dB) shall be allowed over a cumulative bandwidth of 100 Hz, maximum, between 500 and 2,000 Hz. Tolerance levels in terms of dB are defined as:

$$dB = 10 \log \frac{W_1}{W_o}$$

Where W_1 = measured acceleration power spectral density of g^2/Hz units.

The term W_o defines the specified level in g^2/Hz units.

Confirmation of these tolerances shall be made by use of an analysis system providing statistical accuracies corresponding to a bandwidth-time constant product. BT = 50, minimum. Specific analyzer characteristics shall be as specified below or equivalent.

a. On-line, contiguous filter, equalization/analysis system having a bandwidth = B = 50 Hz maximum.

b. Swept frequency analysis systems characterized as follows:

(1.) Constant bandwidth analyzer.

(a) Filter bandwidth as follows:

B = 20 Hz, maximum between 20 and 200 Hz.

B = 50 Hz, maximum between 200 and 2,000 Hz.

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(b) Analyzer averaging time = T = 2 RC = 1 second, minimum, where T = true averaging time and RC = analyzer time constant.

(c) Analysis sweep rate (linear) = S = $\frac{B}{4RC}$ or $\frac{B^2}{8}$ Hz/sec, whichever is smaller.

(2.) Constant percentage bandwidth analyzer.

(a) Filter bandwidth = pf_c = one third octave minimum ($0.23f_c$) where p = percentage and f_c = analyzer center frequency.

(b) Analyzer averaging time = T = $\frac{50}{pf_c}$, minimum.

(c) Analysis sweep rate (logarithmic = S = $\frac{pf_c}{4RC}$, or $\frac{(pf_c)^2}{8}$ Hz/second, whichever is smaller.

c. Digital power spectral density analysis systems employing quantization techniques shall provide accuracies corresponding to the above approach.

The random vibration test level shall be a flat spectrum in the range of 20 to 2000 Hz at a PSD level of 0.02 g^2/Hz with an overall test level of 6.3 g's rms. The test time shall be 40 minutes in each of the three mutually perpendicular axes. The analysis shall yield a statistical accuracy of 120 degrees of freedom. The analysis shall be as shown in Table V.

TABLE V. Vibration analysis.

Bandwidth	Analysis filter
20 - 200 Hz	20 Hz filter
200 - 2000 Hz	50 Hz filter

4.6.4.8 Shock. The pressure indicator shall be subjected to the shock test in accordance with MIL-STD-810, Method 516, Procedure I, Figure 516-2, except the acceleration shall be 25 g's and shall reach peak value in not more than 0.600 second and not less than 0.015 second. When subjected to the shock pulse, the pressure indicator inlet port shall be pressurized to 3000 ±100 psig. At the completion of test, the pressure indicator shall be visually inspected for damage and tested in accordance with 4.6.3.2 and 4.6.3.3.

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4.6.5 Service life. The pressure indicator shall be subjected to the following service life test performed hydrostatically:

a. A total of 500 cycles, as follows:

(1.) Pressurize pressure indicator to 225 ±50 psig.

(2.) Increase pressure uniformly to 4000 ±250 psig.

(3.) Exhaust uniformly to 225 ±50 psig.

b. A total of 4000 cycles, as follows:

(1.) Pressurize pressure indicator to 2500 ±150 psig.

(2.) Increase pressure uniformly to 3500 ±200 psig.

(3.) Exhaust uniformly to 2500 ±150 psig.

c. A total of 500 cycles, as follows:

(1.) Pressurize pressure indicator to 225 ±50 psig.

(2.) Increase pressure uniformly to 4000 ±250 psig.

(3.) Exhaust uniformly to 225 ±50 psig.

At the completion of the test, the pressure indicator shall be visually inspected for damage and tested in accordance with 4.6.3.2 and 4.6.3.3.

4.6.6 Burst pressure. The pressure indicator shall be pressurized hydrostatically to 6700 ±100 psig for a minimum duration of three minutes. There shall be no indication of rupture or internal leakage.

5. PREPARATION FOR DELIVERY

5.1 Pressure indicator preservation and packaging.

5.1.1 Level C. Each pressure indicator shall be provided with a protective cap free from all contaminants and conforming to MIL-C-5501, Type II, Class A; however, protective caps may be made of molded phenolic, or other suitable material. Pressure indicators shall be individually packaged in a heat-sealable bag conforming to MIL-B-22020, Type I. The pressure indicators shall be packaged free of oil, water and water-vapor and the bag shall be dust-free, lint-free and free from water-vapor when sealed.

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5.2 Packing.

5.2.1 Level C. Pressure indicators shall be packed in accordance with MIL-STD-794.

5.3 Marking. All unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The pressure indicator is used to determine the gas pressure in a nitrogen receiver. The pressure indicator is adaptable to a receiver containing air or nitrogen and is capable of operating within the temperature range of -54°C to +121°C (-65°F to +250°F).

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, revision letter and date of this specification.

b. Government-approved test laboratory (see 3.1.2).

c. First article requirement (see 3.1 and 3.1.2).

d. Option for removal from AS/QVPL (see 3.2.4.3).

e. Option for quality reverification (see 3.2.4.3).

f. Responsibility for inspection (see 4.1).

g. Samples subjected to sampling inspection (see 4.5.2) shall not be considered or accepted as part of the contract or purchase order.

h. Rework and retest requirements and quantity to be submitted for inspection lot samples (see 4.5.2).

i. Tests deleted and altered test sequence (see 4.4, 4.5.2 and footnote 2/ on Table III.

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulations, the data requirements identified below will be developed as specified by an approved Date Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements

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List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
3.2	Test Reports	DI-T-21466A
3.2.1, 3.2.2, 3.2.4 & 3.2.4.2	Approved Parts Source	UDI-R-20331B
3.2.4.1	Test Procedures	DI-T-3702A
3.3	Engineering Change Proposal	DI-E-2037
4.4.1	Report, First Article Test	UDI-T-23790

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The first article should be a first production item. The first article should consist of four pressure indicators. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.4 Contract or purchase order provision. Contract or purchase orders should specify the following provision for first article inspection. "When a contractor is in continuous production of the pressure indicator applicable to this specification from contract to contract, consideration shall be given to waive the first article inspections. However, waiver of first article inspections shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract." The words "purchase order" should be substituted throughout for the word "contract" when applicable.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to previous issue due to the extensiveness of the changes.

Preparing activity:

Navy - AS

(Project 6685-N672)

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